RAMBØLL

PROJECT	PROJECT NO.		
—	—	PREPARED	—
		CHECKED	—
		APPROVED	—

Rectangular Beam Bending Check

REV / DATE
0.1 · 2026-07-19

DRAFT

PURPOSE
Elastic bending stress verification for a simply supported rectangular timber beam under uniform load.

1 Input Parameters

Symbol	Value	Unit	Description
b	200	mm	Section width
h	400	mm	Section depth
L	5000	mm	Span
q	4.5	kN/m	Distributed load (design)
f_m	24	MPa	Design bending strength

2 Calculation

Maximum bending moment

$$M_{max} = \frac{q \cdot \left(\frac{L}{1000}\right)^2}{8}$$

$$= \frac{4.5 \cdot \left(\frac{5000}{1000}\right)^2}{8}$$

$$= \mathbf{14.06}\ \mathrm{kNm}$$

Elastic section modulus

$$W = \frac{b \cdot h^2}{6}$$

$$= \frac{200 \cdot 400^2}{6}$$

$$= \mathbf{5.333e+06}\ \mathrm{mm}^3$$

Maximum bending stress

$$\sigma_m = M_{max} \cdot 1000000 \cdot \frac{1}{W}$$

$$= 14.06 \cdot 1000000 \cdot \frac{1}{5.333e+06}$$

$$= \mathbf{2.637}\ \mathrm{MPa}$$

3 Verification

OK $\sigma_m = 2.637\ \mathrm{MPa}\ \leq\ f_m = 24\ \mathrm{MPa}$